

A WITNESS-CORE REDUCTION FOR THE RANK-FOUR, COVERING-NUMBER-THREE CASE OF CHVÁTAL'S CONJECTURE

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ABSTRACT. We study the first rank not covered by the known small-rank theorem in Chvátal's downset conjecture, under the additional hypothesis that the intersecting family has covering number three. We prove an exact witness-core reduction: after fixing a minimum three-cover and three minimality witnesses, a core of at most ten points meets every member in at least two points. Consequently, all exterior residual families have rank at most two. Up to isomorphism there are exactly twenty-nine witness cores, so the original unbounded problem is equivalent to twenty-nine colored rank-two exterior problems. We give the exact section formulas for the generated downset, prove a useful rigidity lemma for large cross-intersecting rank-two families, and provide a sound-and-complete finite Boolean encoding. Independent CP-SAT and Z3 experiments are reported with explicit provenance and with a clear separation between solver output and proof certificates. The main rank-four, covering-number-three case remains open: the present paper is a rigorous reduction and computational audit, not a claimed resolution.

1. INTRODUCTION

A family $\mathcal{D} \subseteq 2^V$ is a *downset* if $A \in \mathcal{D}$ and $B \subseteq A$ imply $B \in \mathcal{D}$. A subfamily $\mathcal{F} \subseteq \mathcal{D}$ is *intersecting* if any two of its members meet. For $v \in V$, write

$$\mathcal{D}(v) = \{A \in \mathcal{D} : v \in A\}.$$

Chvátal's conjecture asserts that

$$(1) \quad |\mathcal{F}| \leq \max_{v \in V} |\mathcal{D}(v)|$$

for every downset and every intersecting subfamily; see [Chv74]. The conjecture is known when $\text{rk}(\mathcal{D}) \leq 3$ [CHK17]. It is also known, without a rank restriction, when the covering number of $\mathcal{F}$ is at most two [FK23].

The first unresolved rank is therefore four. This paper focuses on the next covering-number case

$$\text{rk}(\mathcal{D}) \leq 4, \quad \tau(\mathcal{F}) = 3.$$

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The finite-ground-set frontier has recently advanced: a partitioned VIPR computation verifies the full conjecture for $|V| = 8$ [Loo+26]. That result does not settle the present statement for arbitrary V , and it makes an infinite-to-finite structural reduction, rather than another small- n check, the essential objective.

Our principal result is such a reduction. Three witnesses for the minimality of a fixed cover produce a core H of size at most ten. Every member of $\mathcal{F}$ uses at least two points of H , leaving an exterior residual of size at most two. The witness cores have only twenty-nine isomorphism types. Section 3 turns this observation into exact formulas that preserve both sides of (1).

We emphasize the scope. The reduction is rigorous and infinite, but the final twenty-nine colored rank-two problems have not all been solved. All computational statements below are labelled according to their actual certificate strength.

2. PRELIMINARY REDUCTIONS

For a family $\mathcal{A}$, let

$$\mathcal{A}^\downarrow = \{B : B \subseteq A \text{ for some } A \in \mathcal{A}\}.$$

Lemma 2.1 (minimal generated downset). *If $(\mathcal{D}, \mathcal{F})$ violates (1), then $(\mathcal{F}^\downarrow, \mathcal{F})$ also violates it. Thus it is enough to consider $\mathcal{D} = \mathcal{F}^\downarrow$.*

Proof. Since $\mathcal{D}$ is a downset containing $\mathcal{F}$, one has $\mathcal{F}^\downarrow \subseteq \mathcal{D}$. Every star can only decrease when $\mathcal{D}$ is replaced by $\mathcal{F}^\downarrow$, while $|\mathcal{F}|$ is unchanged. $\square$

Lemma 2.2 (no small members). *If $\mathcal{F}$ is intersecting and $\tau(\mathcal{F}) = 3$, then every member of $\mathcal{F}$ has size at least three.*

Proof. Every member of an intersecting family is itself a cover. A member of size at most two would therefore give $\tau(\mathcal{F}) \leq 2$. $\square$

The next theorem is the main structural contribution.

Theorem 2.3 (witness-core reduction). *Let $\mathcal{F}$ be intersecting, let $\tau(\mathcal{F}) = 3$, and suppose that every member has size at most four. Fix a minimum cover $T = \{1, 2, 3\}$. There exist $A_1, A_2, A_3 \in \mathcal{F}$ such that*

$$A_i \cap T = \{i\} \quad (i = 1, 2, 3).$$

For

$$H = T \cup A_1 \cup A_2 \cup A_3$$

one has $|H| \leq 10$, and every $A \in \mathcal{F}$ satisfies $|A \cap H| \geq 2$. In particular,

$$|A \setminus H| \leq 2 \quad (A \in \mathcal{F}).$$

Proof. For each i , the two-set $T \setminus \{i\}$ is not a cover. Choose $A_i \in \mathcal{F}$ disjoint from it. Since T covers $\mathcal{F}$, the required trace identity follows.

Put $X_i = A_i \setminus T$. Lemma 2.2 and the rank bound give $2 \leq |X_i| \leq 3$. For $i \neq j$, the sets A_i and A_j intersect; their traces on T are disjoint, so $X_i \cap X_j \neq \emptyset$. Write $q = |X_1 \cap X_2 \cap X_3|$. If $q = 0$, inclusion-exclusion gives

$$|X_1 \cup X_2 \cup X_3| \leq 9 - 3 = 6.$$

If $q \geq 1$, each pairwise intersection has size at least q , and

$$|X_1 \cup X_2 \cup X_3| \leq 9 - 3q + q = 9 - 2q \leq 7.$$

Thus $|H| \leq 3 + 7 = 10$.

Every $A \in \mathcal{F}$ meets T and each A_i . If $A \cap H$ were a singleton $\{z\}$, then $z \in T$, say $z = i$. But $i \notin A_j$ for $j \neq i$, contradicting $A \cap A_j \neq \emptyset$. Hence $|A \cap H| \geq 2$, and the last assertion follows from $|A| \leq 4$. $\square$

The bound ten is attained at the level of witness triples: take $|X_i| = 3$ with a common point and otherwise disjoint petals of size two. It is important that Theorem 2.3 bounds a witness core, not the whole support of $\mathcal{F}$.

Proposition 2.4 (a sharp equality family). *Let $V = [5]$, let*

$$\mathcal{D} = \binom{[5]}{\leq 4}, \quad \mathcal{F} = \binom{[5]}{3} \cup \binom{[5]}{4}.$$

Then $\mathcal{F}$ is intersecting, $\tau(\mathcal{F}) = 3$, and

$$|\mathcal{F}| = 15 = |\mathcal{D}(v)| \quad (v \in [5]).$$

Proof. Any two sets of size at least three in a five-set intersect. No two-set covers all three-subsets, whereas every three-set does. Finally, $|\mathcal{F}| = \binom{5}{3} + \binom{5}{4} = 15$ and $|\mathcal{D}(v)| = \sum_{j=0}^3 \binom{4}{j} = 15$. $\square$

3. TWENTY-NINE COLORED RANK-TWO TEMPLATES

Fix witnesses as in Theorem 2.3. Every point of $X_1 \cup X_2 \cup X_3$ is determined, up to relabelling, by the nonempty subset of $[3]$ indexing the X_i that contain it. Let c_J be the multiplicity of membership pattern $\emptyset \neq J \subseteq [3]$. The constraints are

$$(2) \quad 2 \leq \sum_{J \ni i} c_J \leq 3 \quad (i \in [3]),$$

$$(3) \quad \sum_{J \supseteq \{i,j\}} c_J \geq 1 \quad (1 \leq i < j \leq 3).$$

The group S_3 permutes the seven coordinates.

Proposition 3.1 (exact witness-type count). *The nonnegative integer solutions of (2)–(3), modulo the action of S_3 , form exactly twenty-nine orbits. Their core sizes are distributed as follows.*

$ H $	5	6	7	8	9	10
number of types	1	6	10	8	3	1

Proof. Each row sum is at most three, so every c_J lies in $\{0, 1, 2, 3\}$. Exhaust the $4^7 = 16384$ vectors, retain those satisfying (2)–(3), and replace each vector by the lexicographically least of its six S_3 -images. Direct integer enumeration yields the stated orbit and size counts. The accompanying script performs precisely these operations, without floating-point arithmetic or a solver. $\square$

We now state the exact residual formulation. For a fixed witness core let $U = V \setminus H$. Let $\mathcal{Q}$ consist of the subsets $Q \subseteq H$ satisfying

$$2 \leq |Q| \leq 4, \text{quad}Q \cap T \neq \emptyset, \text{quad}Q \cap A_i \neq \emptyset \quad (i = 1, 2, 3).$$

For $Q \in \mathcal{Q}$, define the colored exterior family

$$\mathcal{G}_Q = \{A \setminus H : A \in \mathcal{F}, A \cap H = Q\}.$$

Then every member of $\mathcal{G}_Q$ has size at most $4 - |Q| \leq 2$.

For $P \subseteq H$, put

$$(4) \quad \mathcal{K}_P = \left(\bigcup_{\substack{Q \in \mathcal{Q} \\ P \subseteq Q}} \mathcal{G}_Q \right)^\downarrow \subseteq 2^U.$$

Theorem 3.2 (exact template equivalence). *For every family in Theorem 2.3,*

$$(5) \quad |\mathcal{F}| = \sum_{Q \in \mathcal{Q}} |\mathcal{G}_Q|,$$

$$(6) \quad |\mathcal{D}(h)| = \sum_{\substack{P \subseteq H \\ h \in P}} |\mathcal{K}_P| \quad (h \in H),$$

$$(7) \quad |\mathcal{D}(u)| = \sum_{P \subseteq H} |\mathcal{K}_P(u)| \quad (u \in U),$$

where $\mathcal{D} = \mathcal{F}^\downarrow$. If $Q \cap R = \emptyset$, then $\mathcal{G}_Q$ and $\mathcal{G}_R$ are cross-intersecting. Conversely, colored families satisfying the indicated rank conditions, the cross-intersection conditions, the three witness-containment conditions $\emptyset \in \mathcal{G}_{A_i}$, and $\tau(\mathcal{F}) = 3$ reconstruct exactly a family in the original problem. Hence the rank-four, covering-number-three case is equivalent to the twenty-nine colored rank-two templates.

Proof. The trace classes are disjoint, which gives (5). A set $P \cup R$ with $P \subseteq H$ and $R \subseteq U$ belongs to $\mathcal{F}^\downarrow$ if and only if there are $Q \supseteq P$ and $R_0 \in \mathcal{G}_Q$ with $R \subseteq R_0$. By (4), the exact H -trace- P section of $\mathcal{D}$ is $\{P \cup R : R \in \mathcal{K}_P\}$. Summing the disjoint sections gives (6) and (7).

If $Q \cap R = \emptyset$, intersection of the original sets must occur in U , proving cross-intersection. The converse follows by reversing the construction; the listed conditions preserve rank, intersection, the fixed witnesses, and the covering number. $\square$

The exterior rank bound makes the following elementary rigidity lemma available.

Lemma 3.3 (large cross-intersecting rank-two families). *Let $\mathcal{A}, \mathcal{B} \subseteq \binom{U}{\leq 2}$ be nonempty and cross-intersecting. If $|\mathcal{A}|, |\mathcal{B}| \geq 5$, then some $u \in U$ lies in every member of $\mathcal{A} \cup \mathcal{B}$.*

Proof. Neither family contains the empty set. If $\mathcal{A}$ contains two distinct singletons, then every member of $\mathcal{B}$ contains both points, so $|\mathcal{B}| \leq 1$, a contradiction. If $\{u\} \in \mathcal{A}$, every member of $\mathcal{B}$ contains u . Since $|\mathcal{B}| \geq 5$, a two-set avoiding u cannot meet every member of $\mathcal{B}$; hence every member of $\mathcal{A}$ also contains u .

It remains to suppose that $\mathcal{A}$ consists of two-sets. If it contains two disjoint edges, every member of $\mathcal{B}$ is a two-element transversal of those edges, and there are at most four such transversals. Thus the graph $\mathcal{A}$ has matching number one. A pairwise-intersecting simple graph is a star or is contained in a triangle. Since $|\mathcal{A}| \geq 5$, it is a star with at least five leaves. A set of size at most two meeting all its edges must contain the center, and the result follows. $\square$

In particular, along a connected component of the disjointness graph on trace colors whose residual families all have size at least five, the common exterior point supplied by Lemma 3.3 is the same on every edge. Otherwise a color incident with two different centers would have at most one residual two-set. This observation is a promising bridge from the finite trace templates to exterior-star bounds, but the necessary global weighted inequality is not yet closed.

4. AN EXACT FINITE BOOLEAN MODEL

For fixed n , identify subsets of $[n]$ with bit masks. Introduce Boolean variables y_A and x_S for all $|A|, |S| \leq 4$. The intended meanings are $A \in \mathcal{F}$ and $S \in \mathcal{D}$, respectively. Fix $T = \{1, 2, 3\}$ by relabelling and impose:

- (B1) $y_A = 0$ unless $|A| \in \{3, 4\}$ and $A \cap T \neq \emptyset$;
- (B2) $y_A + y_B \leq 1$ whenever $A \cap B = \emptyset$;
- (B3) for every one- or two-set P , at least one selected A is disjoint from P ;
- (B4) $y_A \leq x_S$ whenever $S \subseteq A$;
- (B5) $x_S \leq \sum_{A \supseteq S} y_A$;
- (B6) for every $i \in [n]$,

$$\sum_A y_A \geq 1 + \sum_{S \ni i} x_S.$$

Proposition 4.1 (soundness and completeness). *The Boolean system (B1)–(B6) is satisfiable if and only if there is a strict counterexample of rank at most four and covering number three on $[n]$ for which T is a minimum cover.*

Proof. Conditions (B1)–(B3) say exactly that the selected family is intersecting, is covered by T , and has no cover of size at most two. Conditions (B4)–(B5) together give

$$x_S = 1 \iff y_A = 1 \text{ for some } A \supseteq S,$$

so x is exactly the minimal generated downset, not merely a containing downset. Finally, (B6) is strict violation of every star. The reverse construction reads any counterexample into the same variables. $\square$

The model is intentionally smaller than a full 2^n encoding: only sets of size at most four occur. Optional constraints can force a particular one of the twenty-nine witness cores, full support, or maximality of the intersecting family.

5. COMPUTATIONAL AUDIT AND FALSIFIED STRENGTHENINGS

The implementation accompanies this manuscript. It contains independent CP-SAT and Z3 encodings and a direct semantic verifier for any returned family. Table 1 records one reproducible CP-SAT regression run of the direct model.

TABLE 1. Direct CP-SAT search for a strict rank-four, $\tau = 3$ counterexample. Times are wall-clock seconds on an Apple Silicon workstation. “UNSAT” here is solver output, not a separately checked proof certificate.

n	variables y	variables x	status	time
5	31	31	UNSAT	0.007
6	57	57	UNSAT	0.175
7	99	99	UNSAT	1.463
8	163	163	UNSAT	42.230

The source SHA-256 for these four manifests is

08259e44b282b3a9e9fd07217f0c31c8f579863763278346fe1ad374f46d107e.

The individual JSON manifests bind the source hash, parameters, solver statistics, platform, and status. The $n = 8$ line is only a local regression check; the independent partitioned VIPR work in [Loo+26] is the relevant published machine-checked finite result. An exploratory full-support $n = 9$ run timed out after 900 seconds and therefore provides no conclusion.

Two tempting strengthenings fail and are useful diagnostics. First, the maximum need not occur at a point of the chosen three-cover. On [6], let

$$\begin{aligned} \mathcal{F} = & \{123\} \\ & \cup \{145, 245, 345, 146, 246, 346, 156, 256, 356\} \\ & \cup \{1456, 2456, 3456\}. \end{aligned}$$

This is an intersecting family with minimum cover $\{1, 2, 3\}$ and $|\mathcal{F}| = 13$. For $\mathcal{D} = \mathcal{F}^\downarrow$, the three cover stars all have size 11, while the stars at 4, 5, 6 have size 16. Thus Chvátal's bound holds, but no cover point witnesses it. Second, the average of the three cover stars can also be smaller than $|\mathcal{F}|$. Any proof must therefore retain exterior or witness-core stars.

For full-support instances, optimization gives the exact best gaps $|\mathcal{F}| - \max_v |\mathcal{D}(v)| = 0, -1, -2$ for $n = 5, 6, 7$, respectively. A gap -3 family was found for $n = 8$, but optimality was not certified in the allotted run. This suggests a deletion-stability inequality, but it is recorded only as a hypothesis.

6. REMAINING BOTTLENECK

Theorem 3.2 isolates the unresolved step. For each of twenty-nine fixed cores, one must prove from the colored rank-two cross-intersection relations that

$$\sum_{Q \in \mathcal{Q}} |\mathcal{G}_Q| \leq \max \left\{ \max_{h \in H} \sum_{P \ni h} |\mathcal{K}_P|, \max_{u \in U} \sum_{P \subseteq H} |\mathcal{K}_P(u)| \right\}.$$

Pairwise downclosure inequalities alone are insufficient: exact integer relaxations admit non-realizable weight vectors. Lemma 3.3 shows why realizability is stronger—large families on disjoint core traces force a common exterior center, and that center creates a large exterior star. A complete proof must make this local rigidity simultaneous across the finite disjointness graph.

The appropriate next certificate is therefore not another unrestricted small- n UNSAT run. It is a finite case theorem for the twenty-nine colored rank-two templates, with each branch justified by the star, triangle, or two-disjoint-edge classification used in Lemma 3.3.

7. REPRODUCIBILITY AND CONCLUSION

The code uses exact Boolean and integer constraints. The witness-type enumerator examines 16384 integer vectors and canonicalizes under all six permutations; its JSON output and source are hash-bound. The search manifests separately record solver-level conclusions. No floating-point optimization, random seed, or unrecorded numerical tolerance enters the structural results.

We have reduced the rank-four, covering-number-three case from an arbitrary ground set to twenty-nine explicit witness cores with rank-two exterior data, and identified the precise global weighted step that remains. This is a substantial narrowing of the problem, but it is not yet a proof of Chvátal's conjecture in the stated class.

APPENDIX A. CANONICAL WITNESS-CORE REPRESENTATIVES

For independent inspection, Table 2 lists every canonical multiplicity vector in the order $(c_1, c_2, c_{12}, c_3, c_{13}, c_{23}, c_{123})$ used by the enumerator. Together with constraints (2)–(3), this table is a compact exact certificate for Proposition 3.1.

TABLE 2. The twenty-nine canonical witness-core representatives.

type	$ H $	multiplicity vector
1	5	(0, 0, 0, 0, 0, 0, 2)
2	6	(0, 0, 1, 0, 1, 1, 0)
3	6	(0, 0, 1, 1, 0, 0, 1)
4	6	(0, 0, 0, 0, 1, 1, 1)
5	6	(0, 0, 0, 1, 0, 0, 2)
6	6	(0, 0, 0, 0, 0, 1, 2)
7	6	(0, 0, 0, 0, 0, 0, 3)
8	7	(1, 1, 0, 1, 0, 0, 1)
9	7	(0, 0, 1, 1, 1, 1, 0)
10	7	(0, 0, 1, 2, 0, 0, 1)
11	7	(0, 1, 0, 1, 1, 0, 1)
12	7	(0, 0, 1, 0, 1, 2, 0)
13	7	(0, 0, 1, 1, 0, 1, 1)
14	7	(0, 1, 0, 1, 0, 0, 2)
15	7	(0, 0, 2, 1, 0, 0, 1)
16	7	(0, 0, 1, 0, 1, 1, 1)
17	7	(0, 0, 1, 1, 0, 0, 2)
18	8	(1, 1, 0, 2, 0, 0, 1)
19	8	(0, 1, 1, 1, 1, 1, 0)
20	8	(0, 1, 1, 2, 0, 0, 1)
21	8	(1, 1, 0, 1, 0, 1, 1)
22	8	(0, 0, 2, 1, 1, 1, 0)
23	8	(0, 0, 2, 2, 0, 0, 1)
24	8	(0, 1, 1, 1, 1, 0, 1)
25	8	(1, 1, 0, 1, 0, 0, 2)
26	9	(1, 2, 0, 2, 0, 0, 1)
27	9	(1, 1, 1, 1, 1, 1, 0)
28	9	(1, 1, 1, 2, 0, 0, 1)
29	10	(2, 2, 0, 2, 0, 0, 1)

The SHA-256 values of the enumerator and its JSON output are, respectively,

f3e00af54c8181c437a620f47f717826eea4e28253dfd883cb85c1eafd256753
f8630641c26014a3ef3359f184801b6e5f8f94eb3279820619f7ea6acd31b59b.

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